山东大学 计算机科学与技术 学院

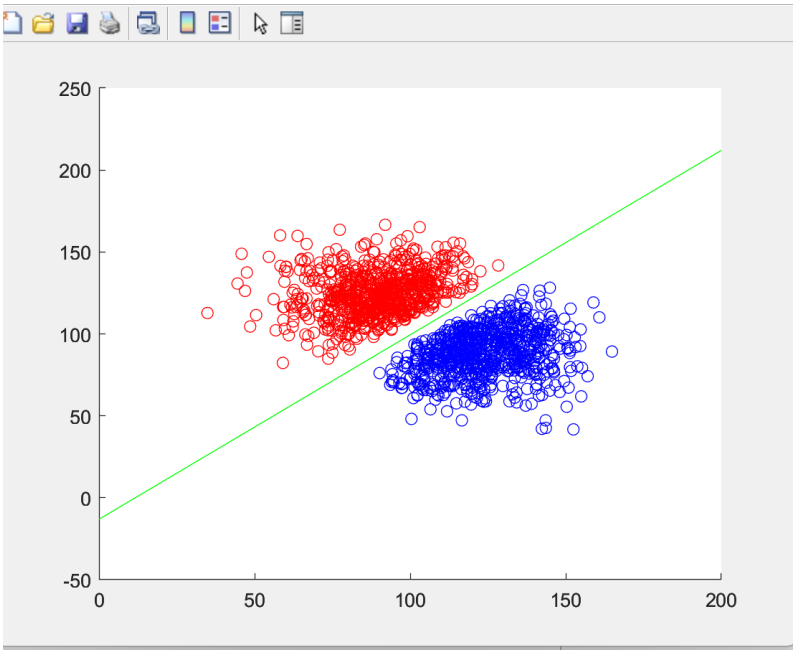
机器学习 课程实验报告

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实验题目：Experiment 5 :SVM		

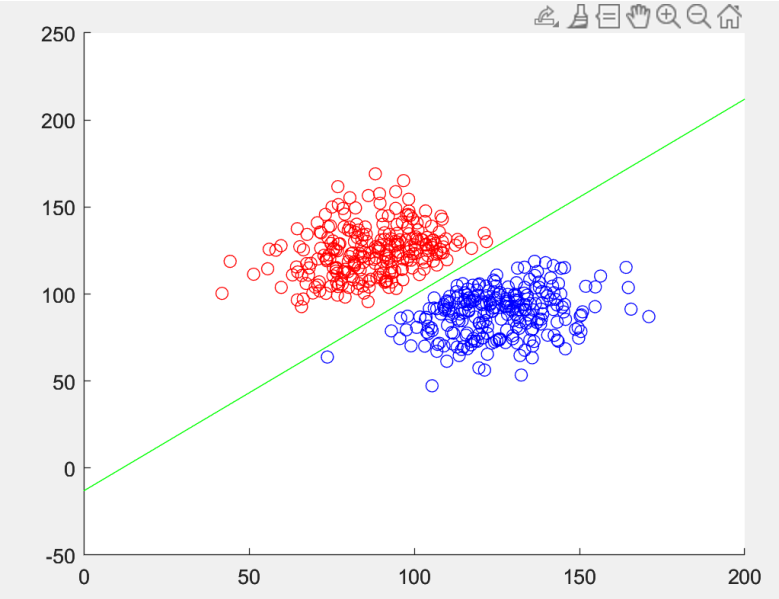
实验过程：

一、SVM

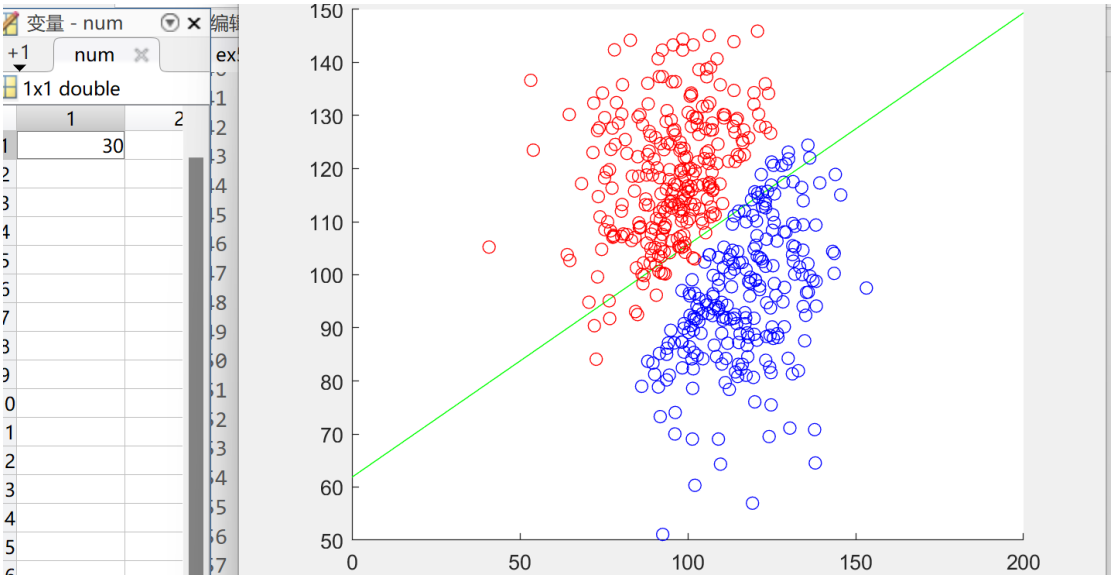
1、首先使用第一个训练集和测试集，此时不使用带有松弛变量的 SVM 模型。训练出模型后，在训练集上绘制模型直线。可以看到所有点都被正确分类。



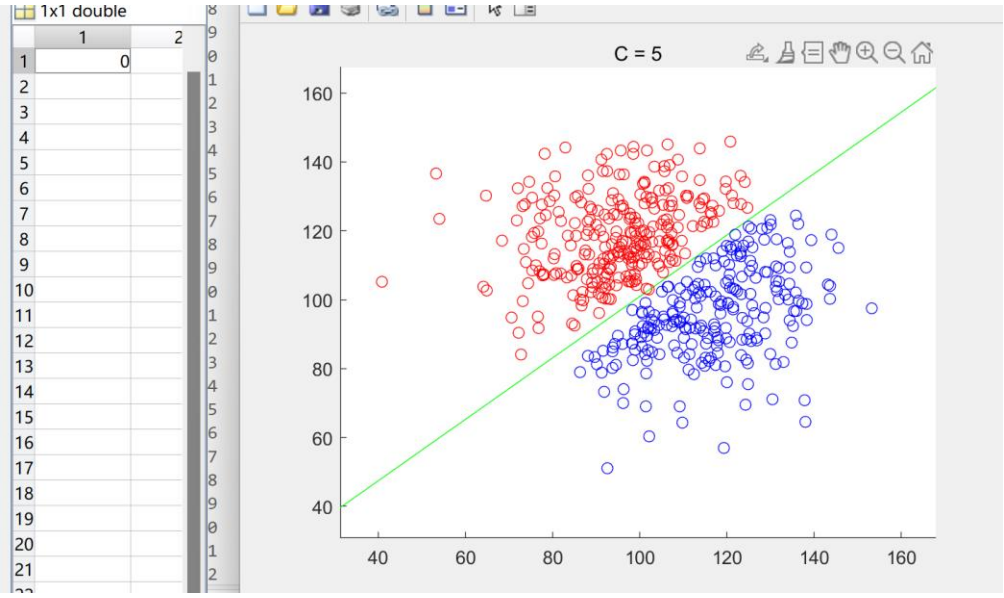
在测试集上绘制模型直线，可以看到，所有点都被正确分类。



2、依旧使用不带有松弛变量的 SVM 模型，训练出模型直线后在 test2 上进行绘制，发现结果如下。可以看到，有许多点被错误分类。且错误率为 30/500 = 6%。



使用带有松弛变量的 SVM 进行模型训练，训练结果如下。可以看到，使用了松弛变量后，错误个数为 0。错误率达到 0%。

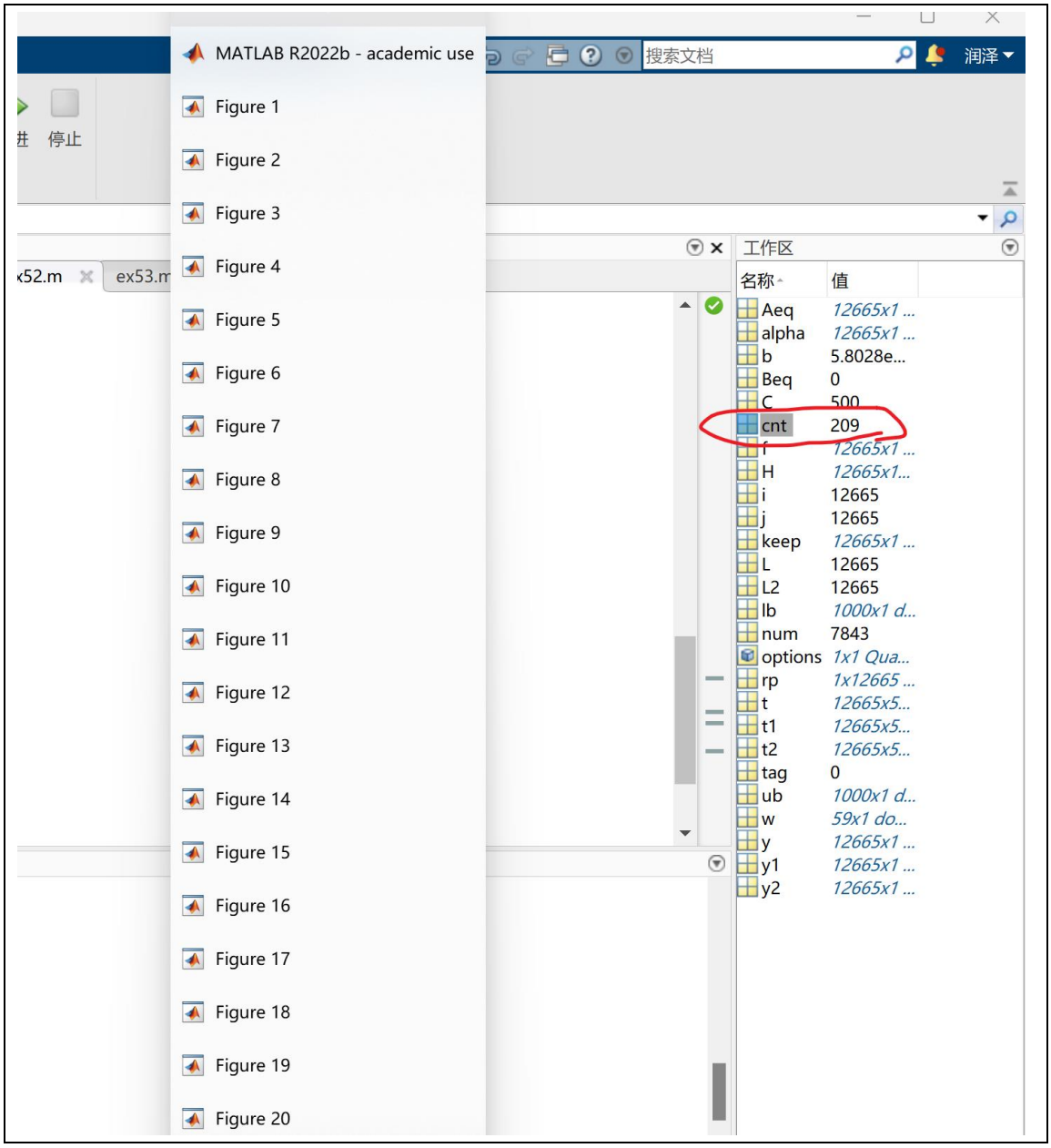


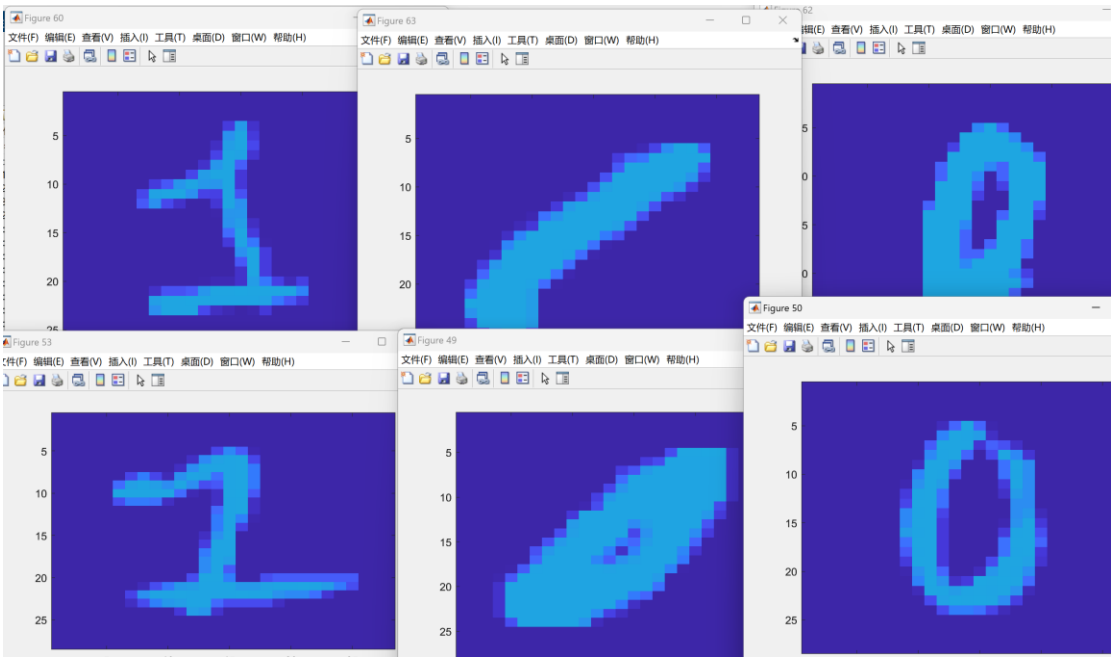
2、Handwritten Digit Recognition

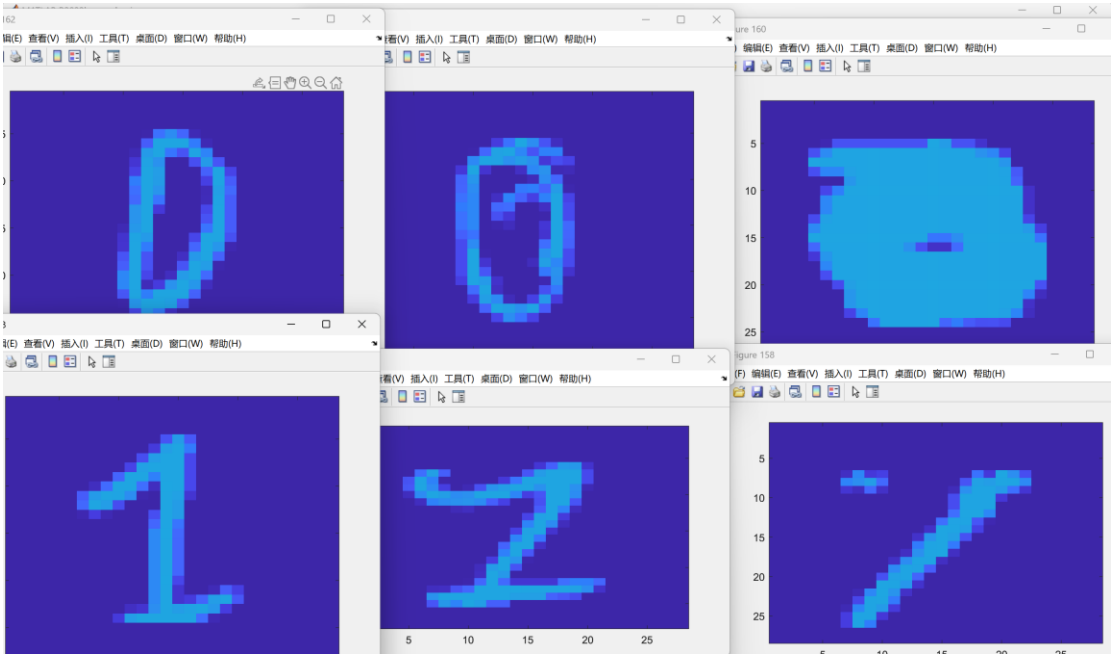
1、没有使用松弛变量的 SVM 模型

如下图，是没有使用松弛变量的 SVM 模型效果

对于训练集来说，有 209 个错误，训练误差为为 1.6502%（虽然 Matlab 变量中 C = 500，但代码中没有用到），任选了几个错误数据图，如第二、三张图

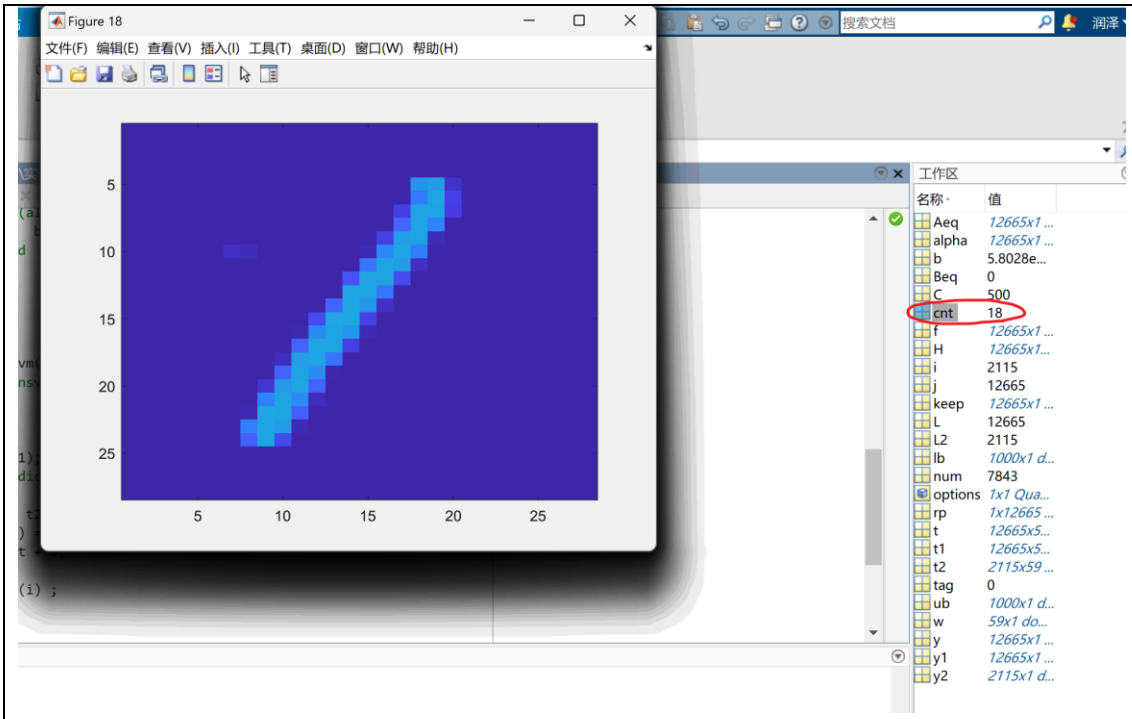

MATLAB R2022b - academic use
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Figure 6
Figure 7
Figure 8
Figure 9
Figure 10
Figure 11
Figure 12
Figure 13
Figure 14
Figure 15
Figure 16
Figure 17
Figure 18
Figure 19
Figure 20
搜索文档
润泽
停止
工作区
名称
值
Aeq 12665x1 ...
alpha 12665x1 ...
b 5.8028e...
Beq 0
C 500
cnt 209
f 12665x1 ...
H 12665x1...
i 12665
j 12665
keep 12665x1 ...
L 12665
L2 12665
lb 1000x1 d...
num 7843
options 1x1 Qua...
rp 1x12665 ...
t 12665x5...
t1 12665x5...
t2 12665x5...
tag 0
ub 1000x1 d...
w 59x1 do...
y 12665x1 ...
y1 12665x1 ...
y2 12665x1 ...





对于测试集

2115 个训练样本，有 18 个错误点，错误率为 0. 851 %



2、使用松弛变量的 SVM

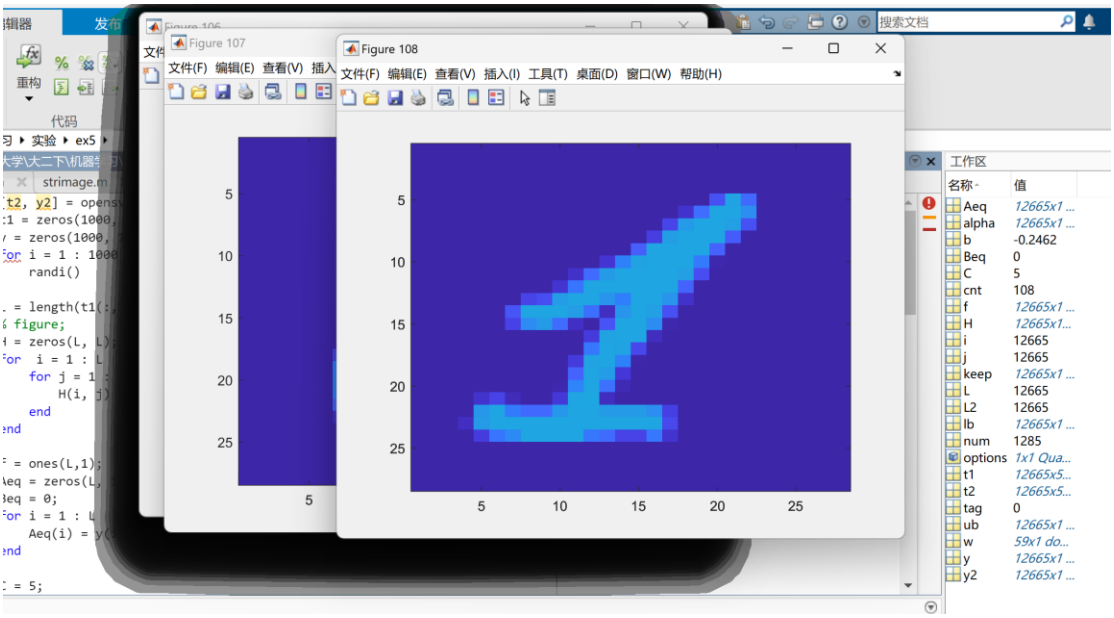
我选择了许多 C 的值，从 1 到 3000 分别为 1、5、20 、30、 50 、100、200、500、1000、3000、5000。结果如下几图：

对于训练集

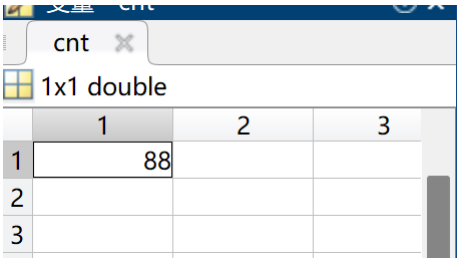
C = 5 时 有 108 个错误 错误率为 108 / 12665 = 0.852%

C = 50 时 有 88 个错误，错误率为 88 /12665 = 0.694%

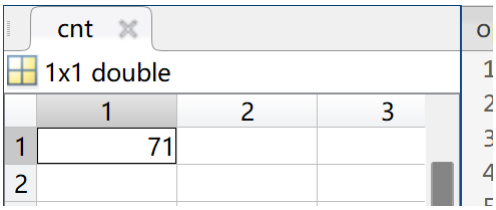
C = 500 时 有 错误率为 71 / 12665 = 0.560%



C = 50



C = 500



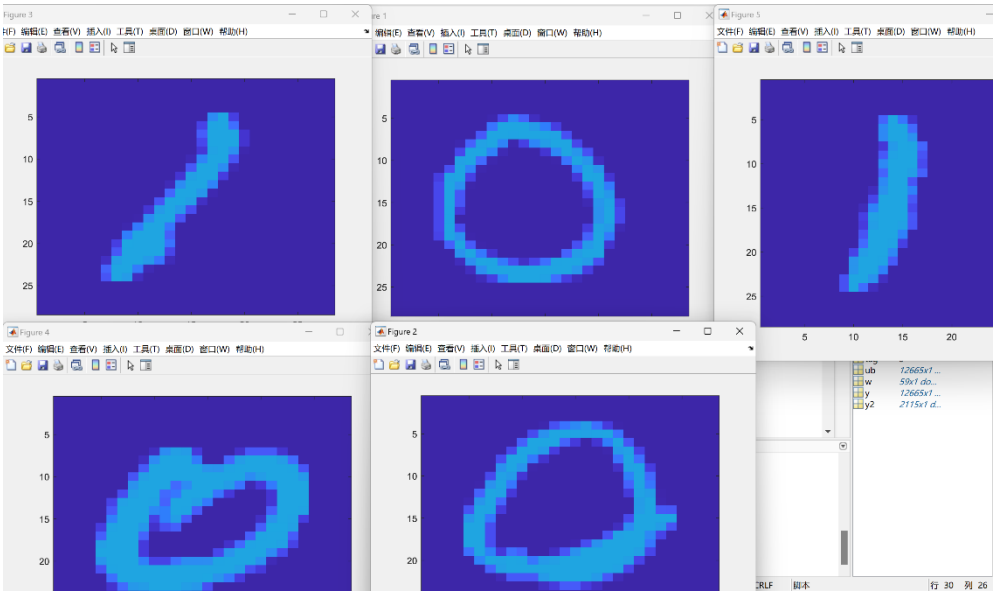
对于测试集

C = 1 时有 5 个错误预测，错误率为 5 / 2115 = 0.236%

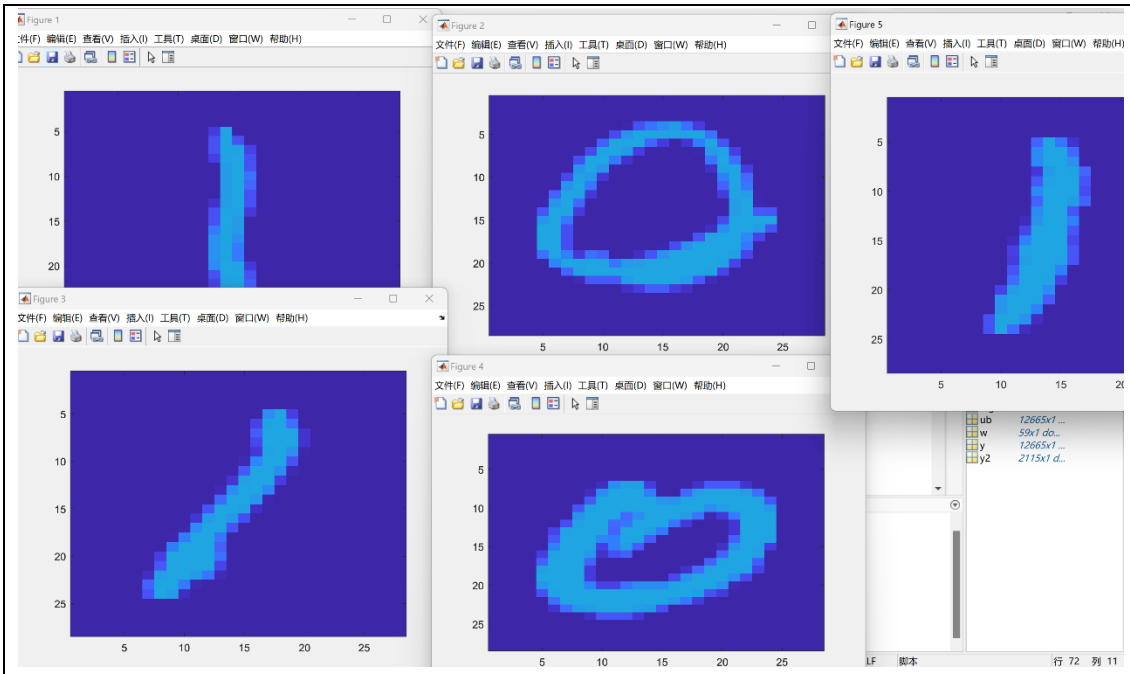
C = 5 到 C = 30 有 4 个错误预测，错误率为 4/ 2115 = 0.189%

C = 30 到 C = 5000，一直是由 2 个错误预测 错误率为 2 / 2115 = 0.0945

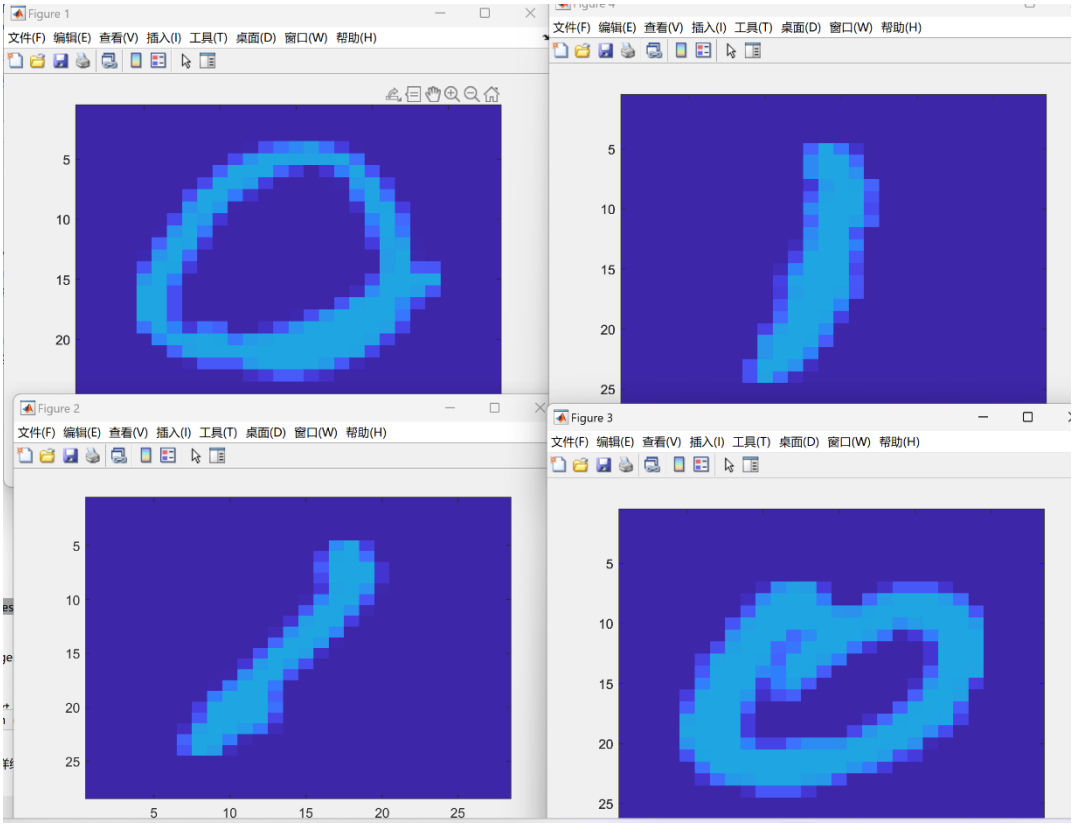
C = 5



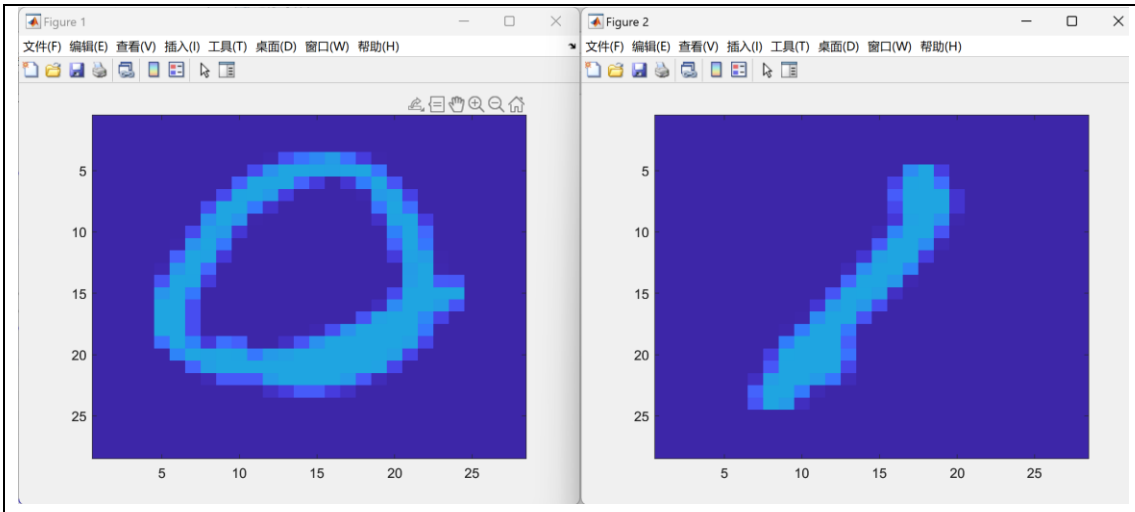
C = 20


Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
文件(F) 编辑(E) 查看(V) 插入(I) 工具(T) 桌面(D) 窗口(W) 帮助(H)
ub 12665x1 ...
w 59x1 do...
y 12665x1 ...
y2 2115x1 d...
LF 脚本 行 72 列 11

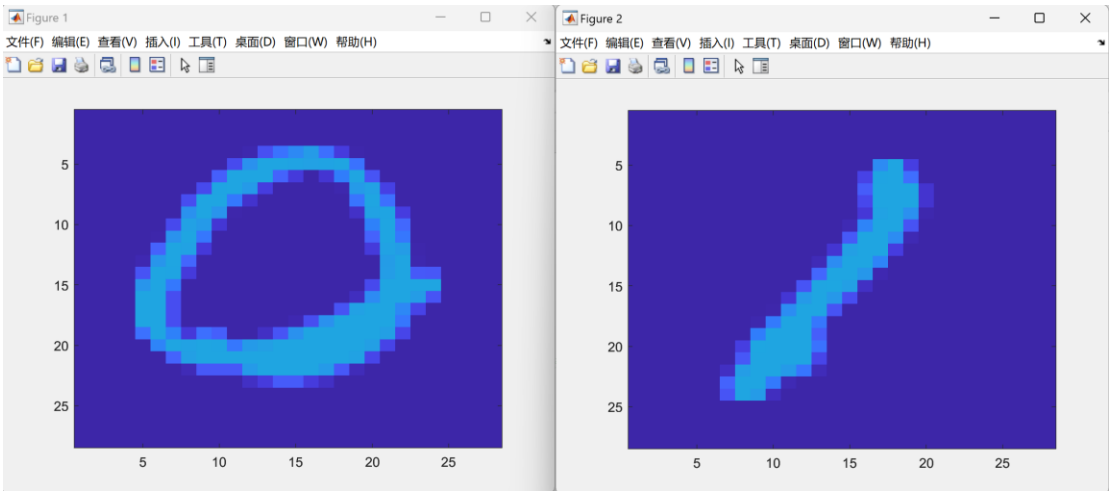
C = 30


Figure 1
Figure 2
Figure 3
文件(F) 编辑(E) 查看(V) 插入(I) 工具(T) 桌面(D) 窗口(W) 帮助(H)

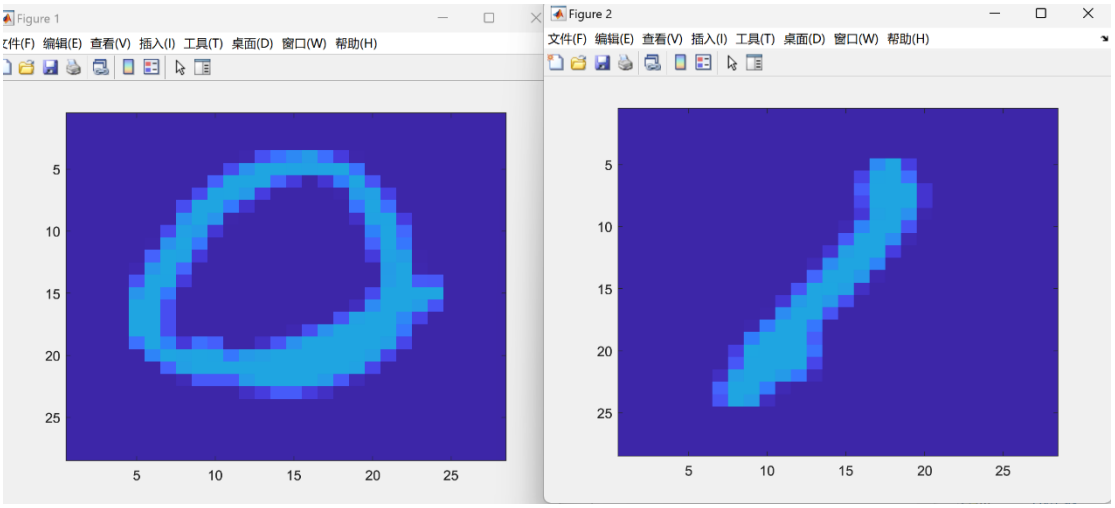
C = 50


Figure 1
文件(F) 编辑(E) 查看(V) 插入(I) 工具(T) 桌面(D) 窗口(W) 帮助(H)
Figure 2
文件(F) 编辑(E) 查看(V) 插入(I) 工具(T) 桌面(D) 窗口(W) 帮助(H)

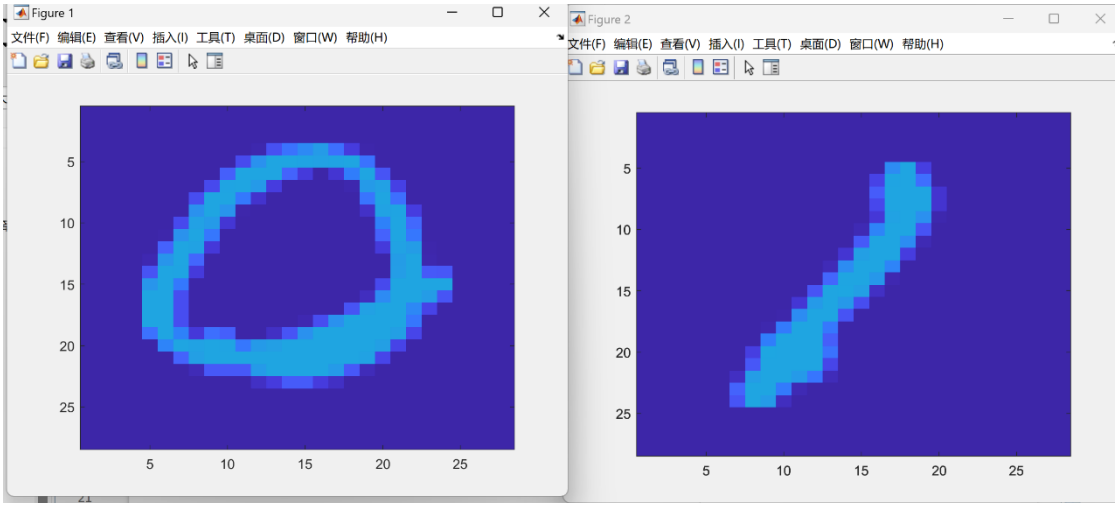
C = 100


Figure 1
文件(F) 编辑(E) 查看(V) 插入(I) 工具(T) 桌面(D) 窗口(W) 帮助(H)
Figure 2
文件(F) 编辑(E) 查看(V) 插入(I) 工具(T) 桌面(D) 窗口(W) 帮助(H)

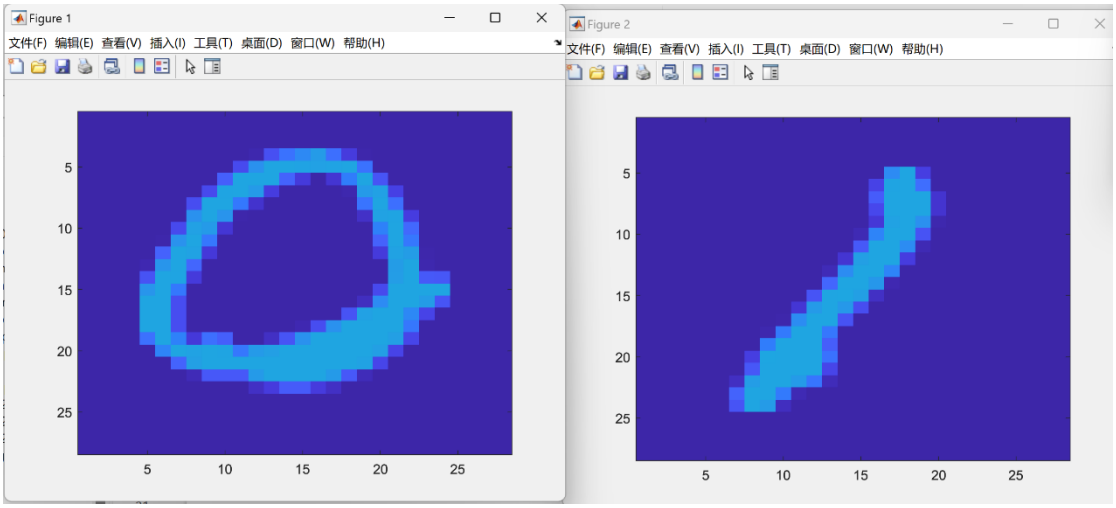
C = 200


Figure 1
Figure 2
文件(F) 编辑(E) 查看(V) 插入(I) 工具(T) 桌面(D) 窗口(W) 帮助(H)

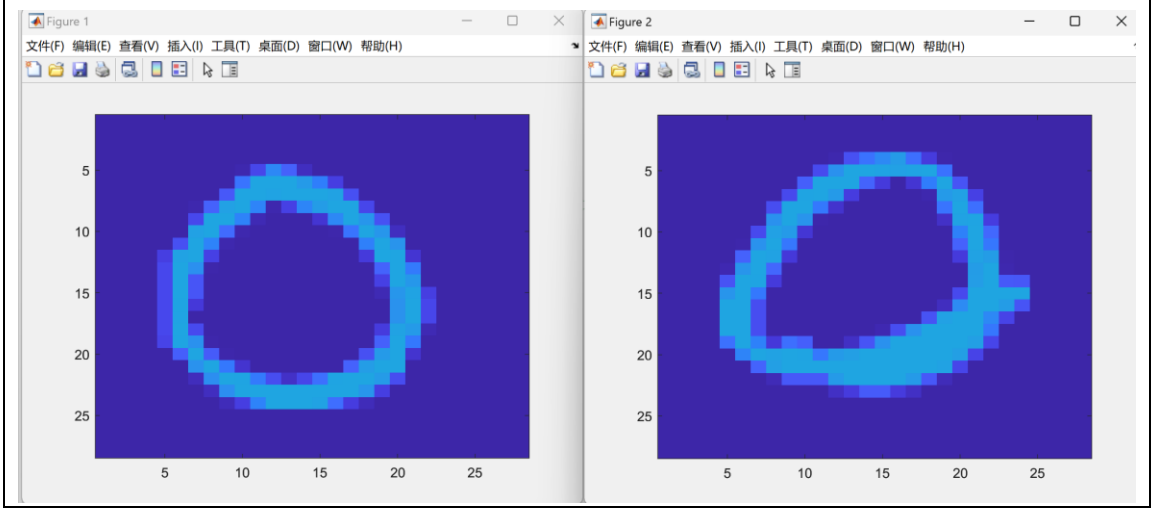
C = 500



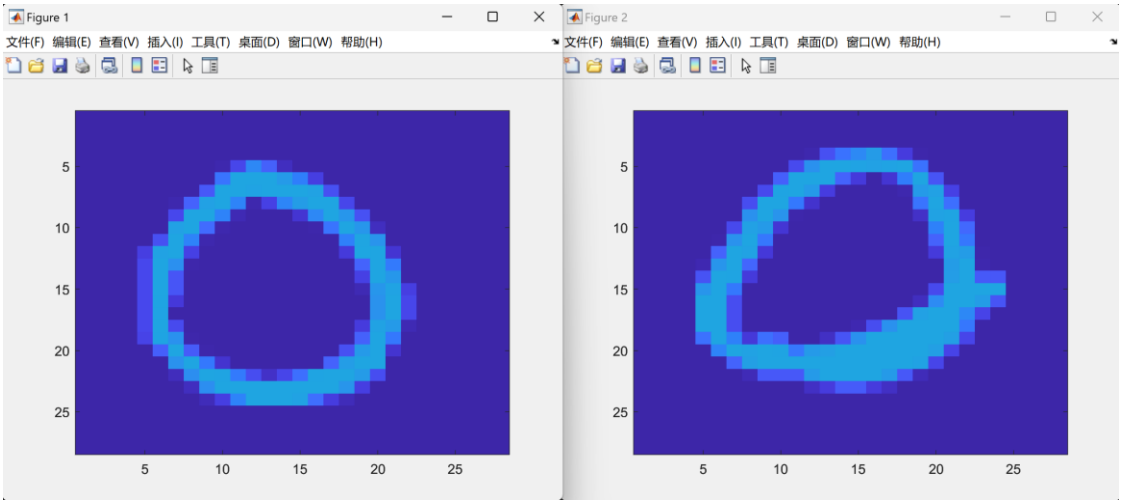
C = 1000



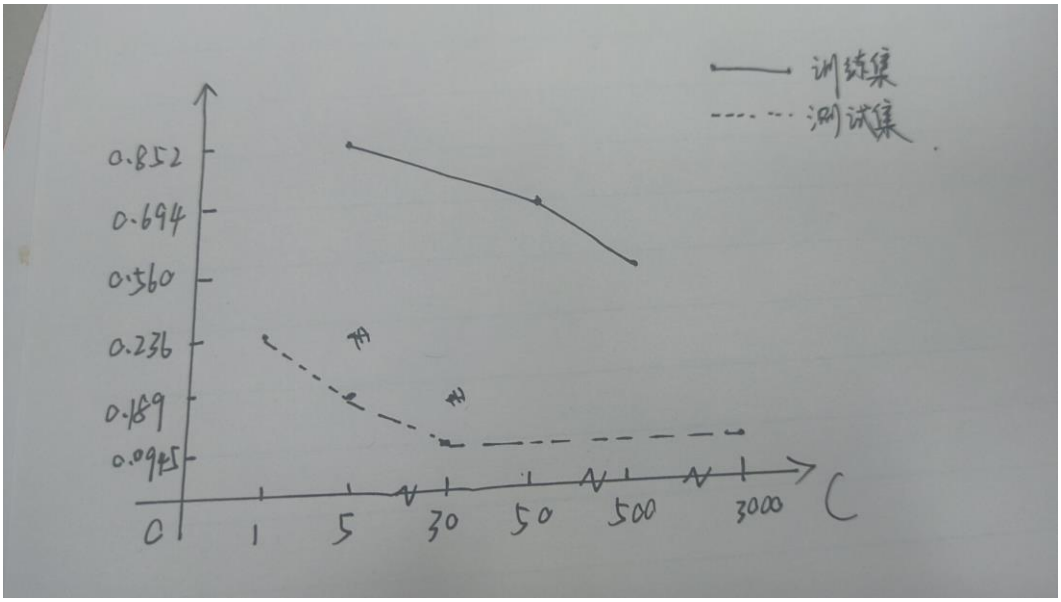
C = 3000



C = 5000



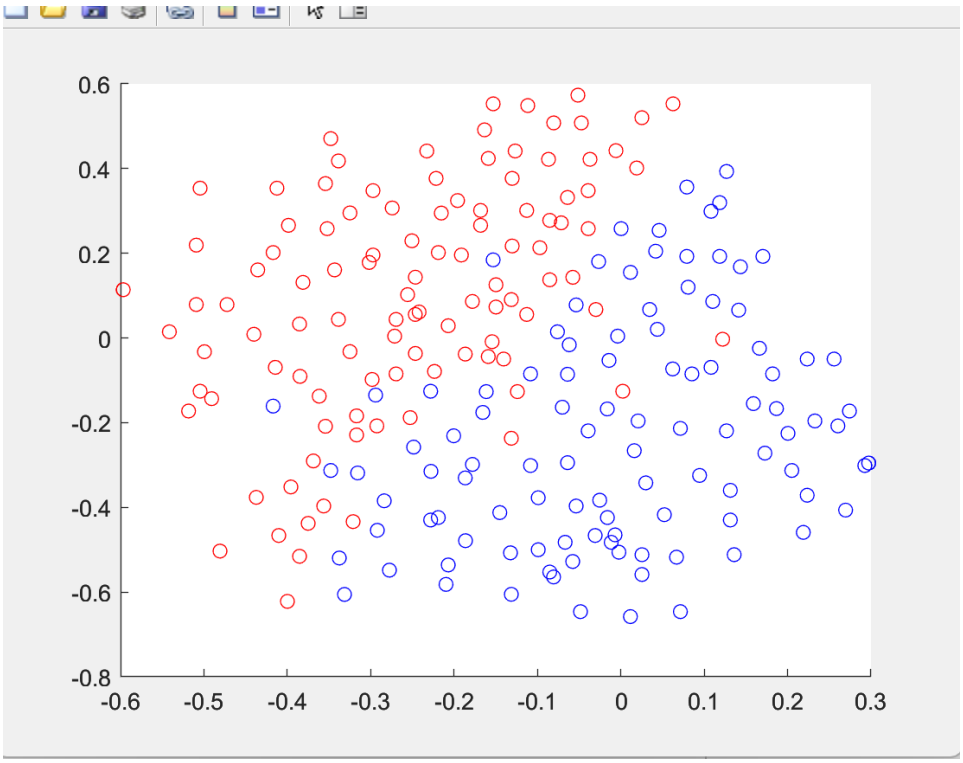
手工绘制图表如下（横坐标为 C 选择，纵坐标为错误率）
得到发现随着 C 增大，训练集和测试集精准率都逐渐增大，最后趋于稳定



（问题：按理说会出现过拟合，但是 C 最大测试到 3000，对于测试集测试来看仍然没有过拟合）

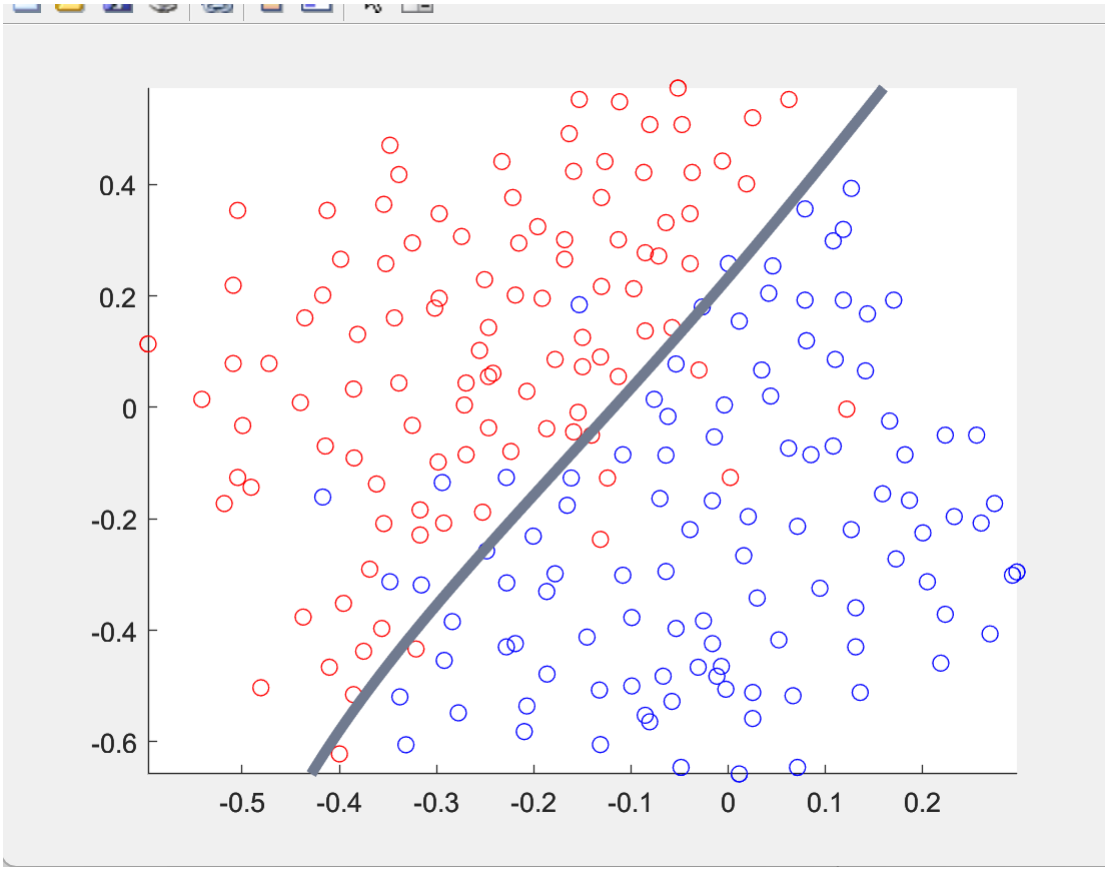
三、核的运用

1、画图可知，不存在线性分类边界

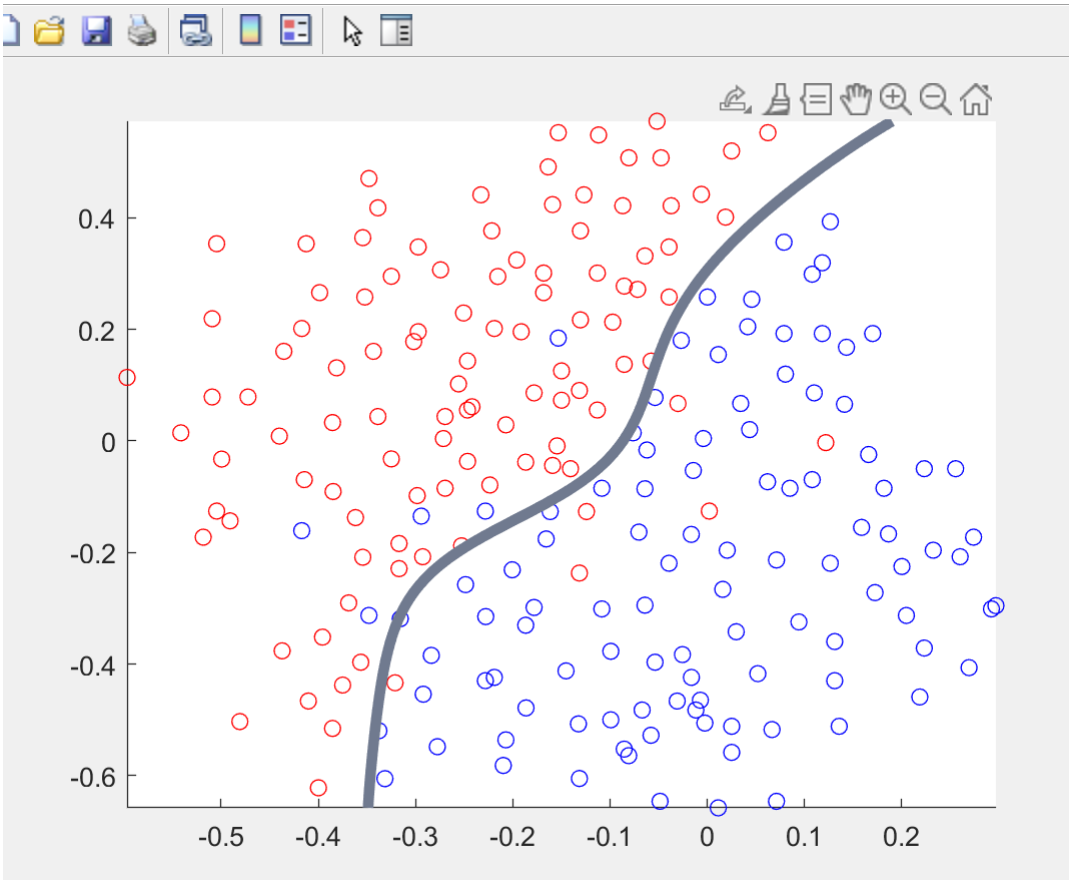

0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-0.6
-0.5
-0.4
-0.3
-0.2
-0.1
0
0.1
0.2
0.3

2、

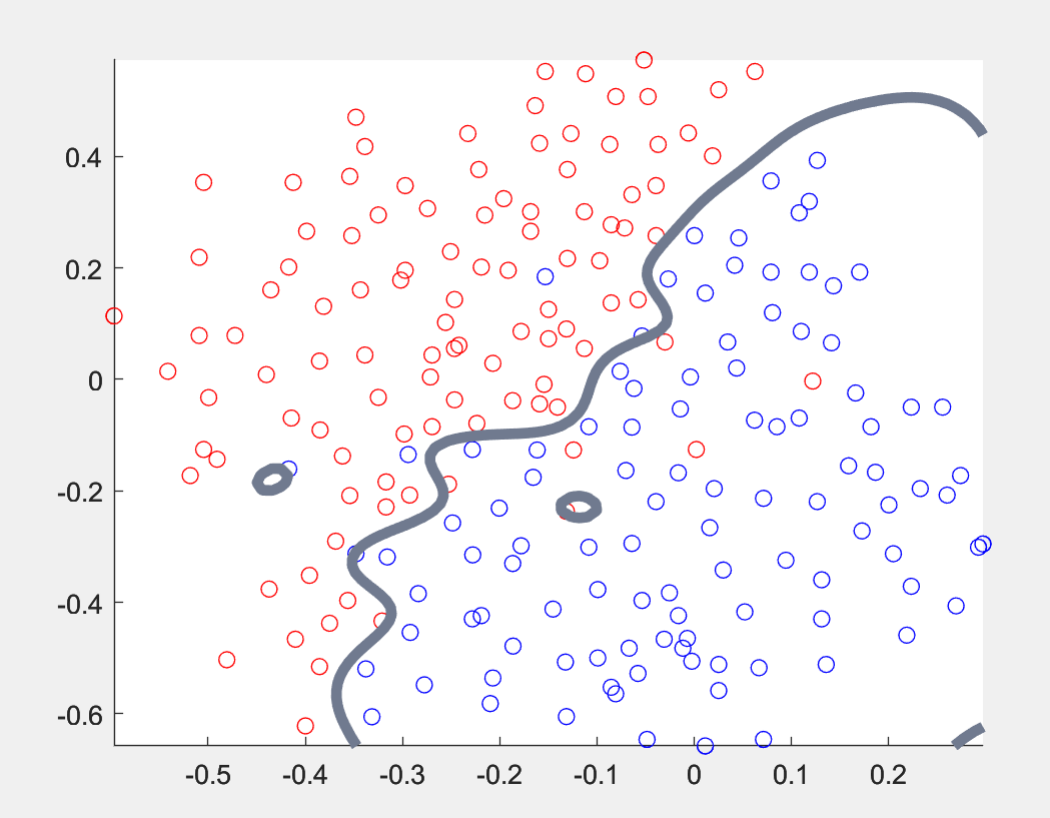
Lamuda = 1


0.4
0.2
0
-0.2
-0.4
-0.6
-0.5
-0.4
-0.3
-0.2
-0.1
0
0.1
0.2

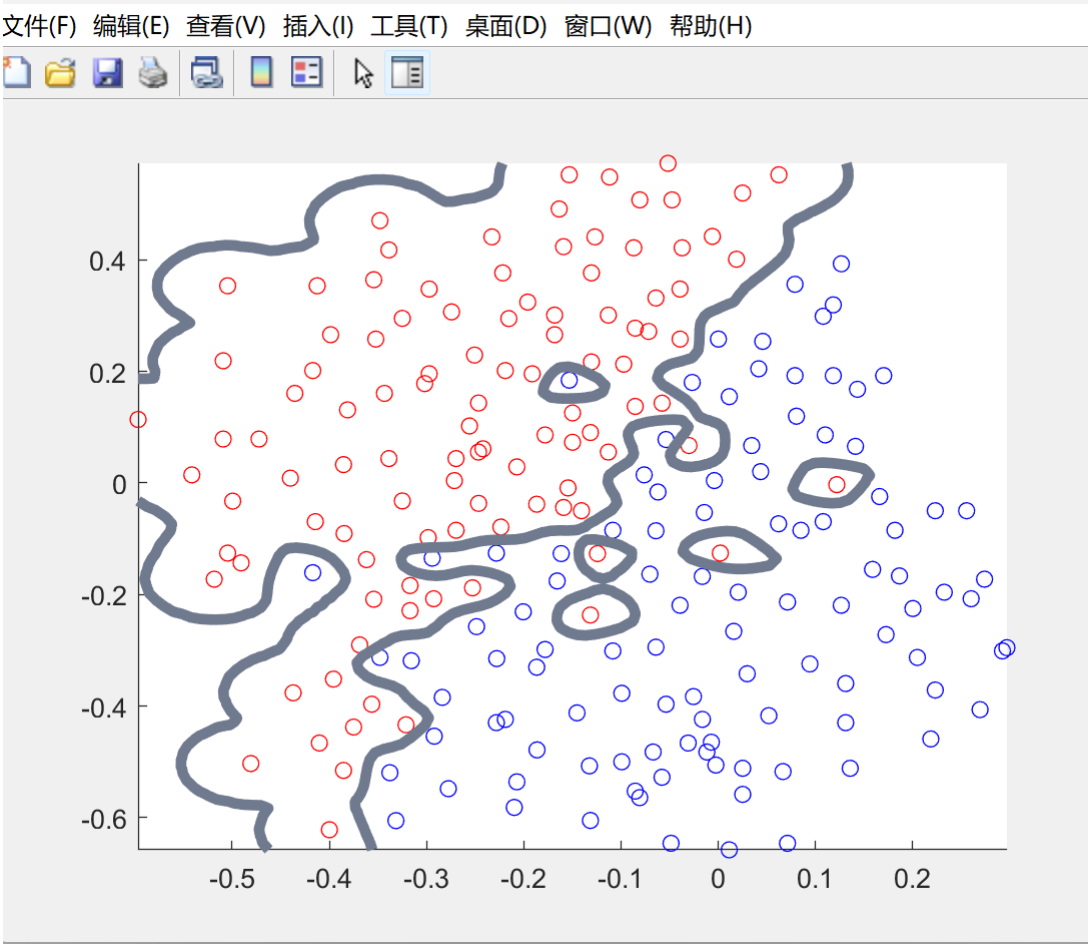
Lamuda = 10



Lamuda = 100



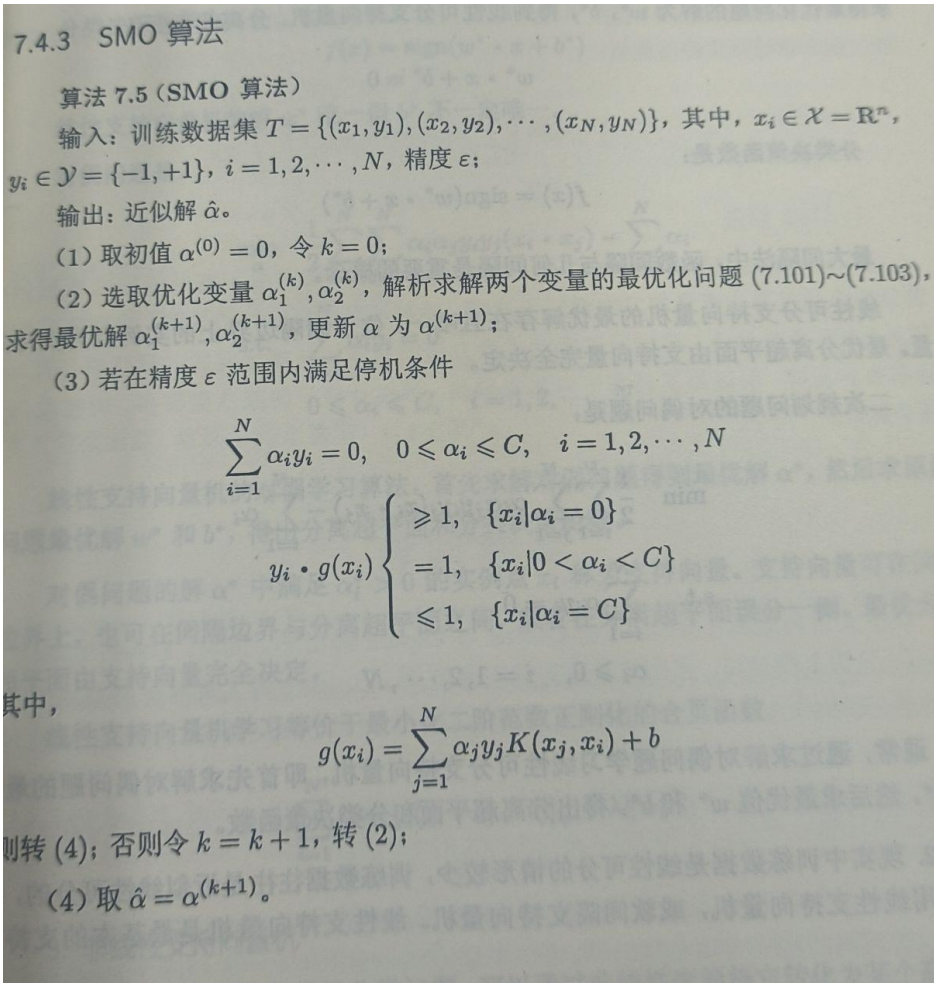
Lamuda = 1000



四、SMO 算法

SMO 算法，即序列最小最优化算法，是一个启发式算法。可以理解为是坐标上升法的更新。在坐标上升法中，我们固定住 n-1 个变量，通过改变 1 个变量的值，使得拉格朗日对偶问题总体达到最小。不断进行这种操作，最后得到一个贪心答案。在 SMO 中，我们固定住 n-2 个变量，只变化 2 个变量。但是不同的是，我们实际只有一个自由度，另一个变量会随着第一个变量的确定而确定，同时满足我们的约束要求。

具体更新方法如下图，最终得到贪心解：



7.4.3 SMO 算法

算法 7.5（SMO 算法）

输入：训练数据集 $T=\{(x_1,y_1),(x_2,y_2),\cdots,(x_N,y_N)\}$，其中，$x_i\in\mathcal{X}=\mathbf{R}^n$，$y_i\in\mathcal{Y}=\{-1,+1\}$，$i=1,2,\cdots,N$，精度 ε；

输出：近似解 $\hat{\alpha}$。

(1) 取初值 $\alpha^{(0)}=0$，令 $k=0$；

(2) 选取优化变量 $\alpha_1^{(k)},\alpha_2^{(k)}$，解析求解两个变量的最优化问题 (7.101)~(7.103)，求得最优解 $\alpha_1^{(k+1)},\alpha_2^{(k+1)}$，更新 α 为 $\alpha^{(k+1)}$；

(3) 若在精度 ε 范围内满足停机条件

$$\sum_{i=1}^{N}\alpha_i y_i=0,\quad 0\leqslant\alpha_i\leqslant C,\quad i=1,2,\cdots,N$$

$$y_i\cdot g(x_i)\begin{cases}\geqslant 1, & \{x_i|\alpha_i=0\}\\ =1, & \{x_i|0<\alpha_i<C\}\\ \leqslant 1, & \{x_i|\alpha_i=C\}\end{cases}$$

其中，

$$g(x_i)=\sum_{j=1}^{N}\alpha_j y_j K(x_j,x_i)+b$$

则转 (4)；否则令 $k=k+1$，转 (2)；

(4) 取 $\hat{\alpha}=\alpha^{(k+1)}$。

代码附录：

模型训练：

%1、SVM（不带松弛变量的模型训练）

```
t1 = readmatrix("training_1.txt");
L = length(t1(:,1));
figure;
hold on;
for i = 1 : L
    if t1(i,3) == 1
        plot(t1(i,1),t1(i,2),'ro');
    else
        plot(t1(i,1),t1(i,2),'bo');
    end
    hold on;
end
hold off;
H = zeros(L, L);
for  i = 1 : L
    for j = 1 : L
```

```
        H(i, j) = H(i, j) + t1(i, 3)*t1(j,3)*(t1(i,1)* t1(j,1)
+t1(i,2)*t1(j,2));
    end
end
f = ones(L,1);
Aeq = zeros(L, 1);
Beq = 0;
for i = 1 : L
    Aeq(i) = t1(i,3);
end
C = 0;
lb = 0; ub = C;
options = optimoptions('quadprog','MaxIterations',1000); % 设置最大迭代次数为
alpha = quadprog(H, -f,[],[],Aeq', Beq,[],[],[],options);
w = [0 ; 0];
for i = 1 : L
    w = w + alpha(i) * t1(i,3) * [t1(i,1); t1(i,2)];
end
typepos = 10000000000;
typeneg = -10000000000;
for i = 1 : L
    if t1(i,3) == 1
        typepos = min(typepos, w(1)*t1(i,1) + w(2)*t1(i,2));
    else
        typeneg = max(typeneg, w(1)*t1(i,1) + w(2)*t1(i,2));
    end
end
hold on;
b = -(typepos + typeneg) / 2;
k = -w(1)/w(2); b2 = -b/w(2);
displayx = 0 : 0.1 :200;
displayy = displayx * k + b2;
plot(displayx, displayy,'g');
hold on;


% 使用了松弛变量的 SVM 求解（对于 traning2.txt and test_2.txt）c = 5
t1 = readmatrix("training_2.txt");
L = length(t1(:,1));
figure;
H = zeros(L, L);
for  i = 1 : L
    for j = 1 : L
```

```
        H(i, j) = H(i, j) + t1(i, 3)*t1(j,3)*(t1(i,1)* t1(j,1)
+t1(i,2)*t1(j,2));
    end
end
f = ones(L,1);
Aeq = zeros(L, 1);
Beq = 0;
for i = 1 : L
    Aeq(i) = t1(i,3);
end
C = 5;
lb = 0; ub = C;
options = optimoptions('quadprog','MaxIterations',1000); % 设置最大迭代次数为
alpha = quadprog(H, -f,[],[],Aeq', Beq,lb,ub,[],options);
% alpha = quadprog(H, -f,[],[],Aeq', Beq,[],[],[],options);
w = [0 ; 0];
for i = 1 : L
    w = w + alpha(i) * t1(i,3) * [t1(i,1); t1(i,2)];
end
typepos = 10000000000;
typeneg = -10000000000;
for i = 1 : L
    if t1(i,3) == 1
        typepos = min(typepos, w(1)*t1(i,1) + w(2)*t1(i,2));
    else
        typeneg = max(typeneg, w(1)*t1(i,1) + w(2)*t1(i,2));
    end
end
hold on;
b = -(typepos + typeneg) / 2;
k = -w(1)/w(2); b2 = -b/w(2);
displayx = 0 : 0.1 :200;
displayy = displayx * k + b2;
plot(displayx, displayy,'g');
title('C = 5')
hold on;
test1 =  readmatrix("test_2.txt");
for i = 1 : length(test1)
    if test1(i,3) == 1
        plot(test1(i,1),test1(i,2),'ro');
    else
        plot(test1(i,1),test1(i,2),'bo');
    end
    hold on;
```

```
end
hold off;
num = 0;
for i = 1 : length(test1)
    if(w(1)*test1(i,1) + w(2)* test1(i,2) + b > 0 && test1(i,3) == -1) num =
num + 1;
    elseif(w(1)*test1(i,1) + w(2)* test1(i,2) + b < 0 && test1(i,3) == 1)
num = num + 1;
    end
end
```

%Handwritten Digit Recognition

%函数 opensvm，读入数据用来返回一个被压缩为列数为 59 的矩阵，简化图片二维矩阵复杂度

```
function [res,sym] = opensvm(filename)
  fidin = fopen(filename); % 打开 filename 文件
  i = 1;
  apres = [];

while ~feof(fidin)
  tline = fgetl(fidin); % 从文件读行
  apres{i} = tline;
  i = i+1;
end
  Length = length(apres);
  res = zeros(Length, 59);
  sym = zeros(Length,1);

  for j = 1 : length(apres)
  a = char(apres(j));
  if a(1) == '-'
      sym(j) = -1;
  else
      sym(j) = 1;
  end
  lena = size(a);
  lena = lena(2);
  xy = sscanf(a(4:lena), '%d:%d');

  lenxy = size(xy);
  lenxy = lenxy(1);
      grid = [];
      grid(784) = 0;
```

```
      for i=2:2:lenxy  %% 隔一个数
          if(xy(i)<=0)
              break
          end
        grid(xy(i-1)) = xy(i) * 100/255;
      end
      grid1 = reshape(grid,28,28);
      grid1 = fliplr(diag(ones(28,1)))*grid1;
      grid1 = rot90(grid1,3);
      x = extractLBPFeatures(grid1);
      for k = 1 : length(x)
           res(j, k) = x(k);
      end
  end
end

% 手写数字识别代码（C = 5，使用测试集测试，并画出不符合要求的图）
[t1, y] = opensvm('train-01-images.svm');
L = length(t1(:,1));
% figure;
H = zeros(L, L);
for  i = 1 : L
    for j = 1 : L
        H(i, j) = y(i) * y(j) * t1(i, :) * t1(j, :)';
    end
end

f = ones(L,1);
Aeq = zeros(L, 1);
Beq = 0;
for i = 1 : L
    Aeq(i) = y(i);
end

C = 5;
lb = zeros(L,1); ub = C * ones(L, 1);
options = optimoptions('quadprog','MaxIterations',500); % 设置最大迭代次数为
alpha = quadprog(H, -f,[], [], Aeq', Beq, lb, ub, [], options);

% alpha = quadprog(H, -f,[],[], Aeq', Beq);

for i = 1 : L
    if alpha(i) < 0.00001
        alpha(i) = 0;
```

```
    end
end

w = zeros(size(t1, 2),1);
for i = 1 : L
    w = w + alpha(i) * y(i) * t1(i,:)';
end

num = 0;
b = 0;
tag = 0;

for i = 1 : L
    if(alpha(i)> 0.001)
        num = num + 1;
        b = b + y(i);
        for j = 1 : L
            if(alpha(j) > 0.001)
                b = b - alpha(j)*y(j) * t1(j,:) * t1(i,:)';
            end
        end
    end
end
b = b / num;

[t2, y2] = opensvm('test-01-images.svm');
% [t2, y2] = opensvm('train-01-images.svm');
L2 = length(t2);
cnt = 1;

keep = zeros(L, 1);
% training repredict wrong about 50%
for i = 1 : L2
    if sign(w' * t2(i,:)' + b) * y2(i) < 0
        keep(cnt) = i;
        cnt = cnt + 1;
        figure;
        strimage(i) ;
    end
end
cnt = cnt - 1;


% 高斯核函数
```

```
function [res] = Kernel(x1, x2, lamuda)
% gauss Kernel
res = exp(-lamuda * (x1 - x2)' *(x1 - x2));
end


%使用核技巧训练线性不可分模型，lamuda = 100
t = readmatrix("training_3.text");

t1 = [t(:, 1) , t(:, 2)];
y = t(: ,3);
L = length(t1(:,1));
figure;
hold on;
for i = 1 : L
    if y(i) == 1
        plot(t1(i, 1), t1(i, 2), 'or');
    elseif y(i) == -1
        plot(t1(i, 1), t1(i, 2), 'ob');
    end
end


lamuda = 100;
H = zeros(L, L);

for  i = 1 : L
    for j = 1 : L
        H(i, j) = y(i) * y(j) * Kernel(t1(i, :)' , t1(j, :)', lamuda);
    end
end

f = ones(L,1);
Aeq = zeros(L, 1);
Beq = 0;
for i = 1 : L
    Aeq(i) = y(i);
end

C = 5;
lb = zeros(L,1); ub = C * ones(L, 1);
options = optimoptions('quadprog','MaxIterations',500); % 设置最大迭代次数为
alpha = quadprog(H, -f,[], [], Aeq', Beq, lb, ub, [], options);
```

```
% alpha = quadprog(H, -f,[],[],Aeq', Beq,[],[],[],options);

for i = 1 : L
    if alpha(i) < 0.00001
        alpha(i) = 0;
    end
end

% Make classificationpredictionsoveragridofvalues
xplot = linspace (min(t1(:, 1)) , max(t1(:, 1)), 100)' ;
yplot = linspace (min(t1(:, 2)), max(t1(:, 2)), 100)' ;
[X, Y] = meshgrid (xplot, yplot) ;
vals= zeros(size(X)) ;

num = 0;
b = 0;
for i = 1 : L
    if(alpha(i)> 0.0001)
        num = num + 1;
        b = b + y(i);
        for j = 1 : L
            if(alpha(j) > 0.001)
                b = b - alpha(j) * y(j) * Kernel(t1(j, :)' , t1(i, :)',
lamuda);
            end
        end
    end
end
b = b / num;

for i = 1 : length(xplot)
    for j = 1 : length(yplot)
        symb = 0;
        for k = 1 : length(alpha)
            if alpha(k) < 0.0001
                continue;
            else
                vals(i, j) = vals(i, j) + alpha(k) * y(k) * Kernel([X(i, j) ;
Y(i, j)], t1(k, :)', lamuda);
                symb = symb + 1;
            end
        end
        if symb ~= 0
```

```
            vals(i, j) = vals(i, j) + b;
        end
    end
end

%
% % For each pointinthisgrid , you need to compute its decision
% % value . Store the decision values invals .
% % Plot the SVM boundary
colormap bone
contour (X, Y, vals, [0 0], 'LineWidth'  , 4)
```

结果分析与体会：

结果分析：

在 part2，手写数字识别的实验中，我的模型没有随着 C 的增大产生过拟合线性，不清楚是不是 C 还不够大。理论上来说，C 应该取一个合适值，而不是越大越好。

体会

SVM 实验是到目前为止我做过的实验中最难的一个实验。难主要体现在内容量大，内容难度高。我深刻理解了怎样通过拉格朗日乘子和拉格朗日对偶性来转化求解一个二次规划问题。